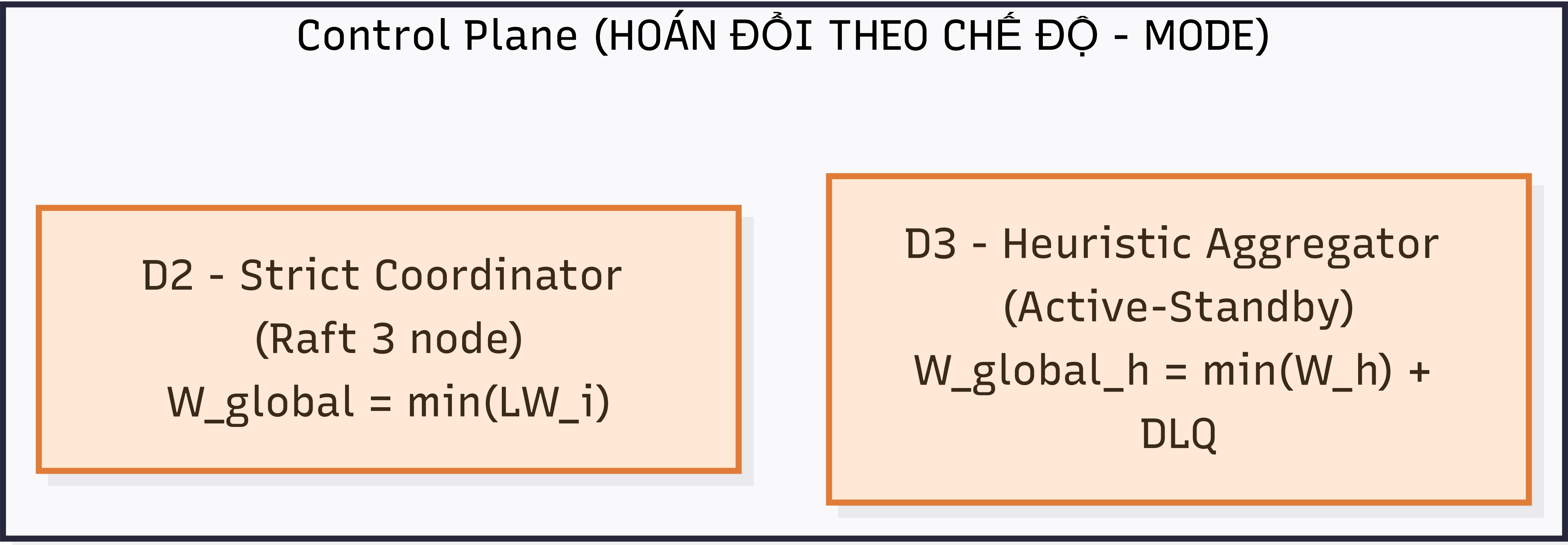
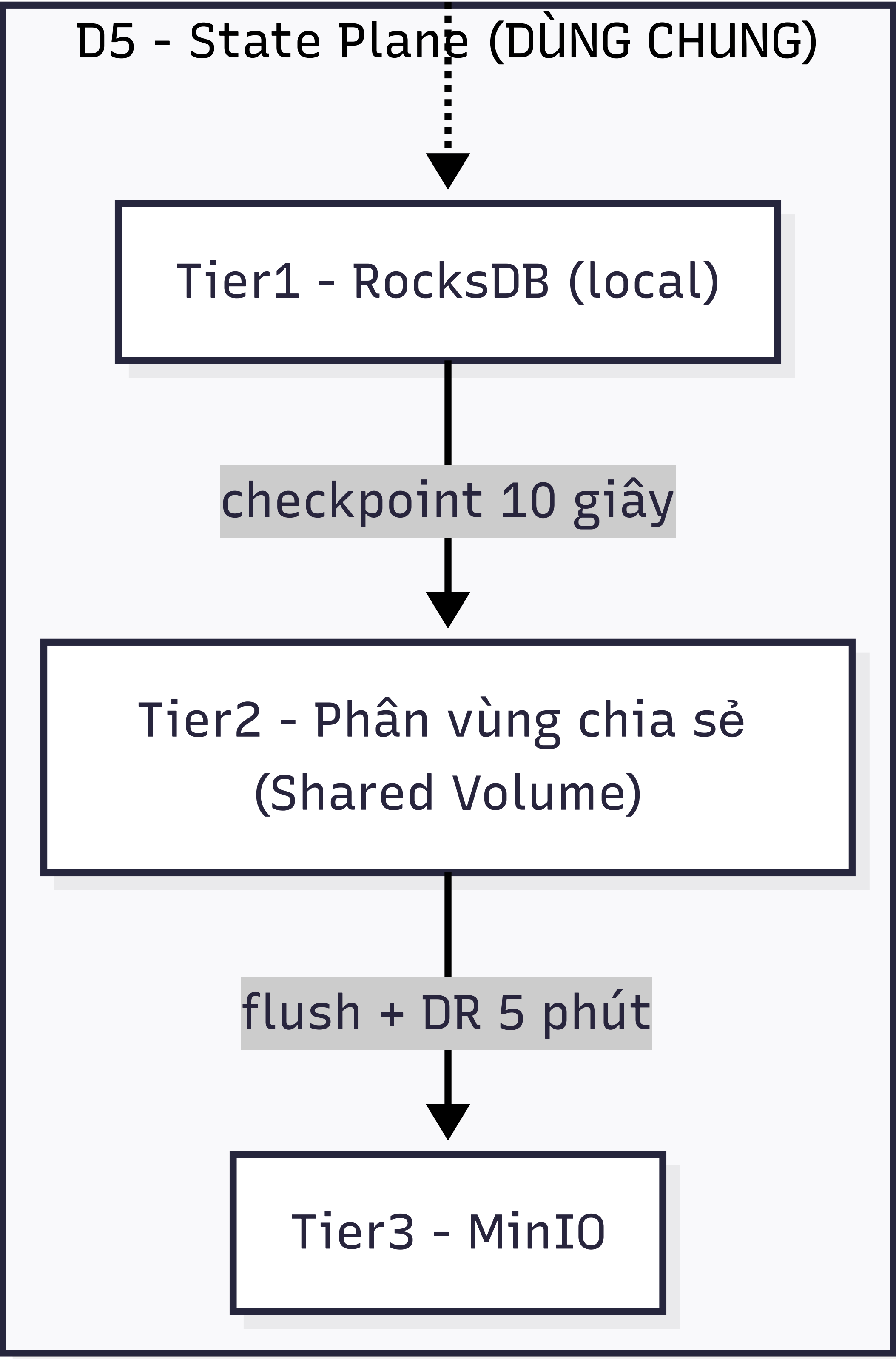
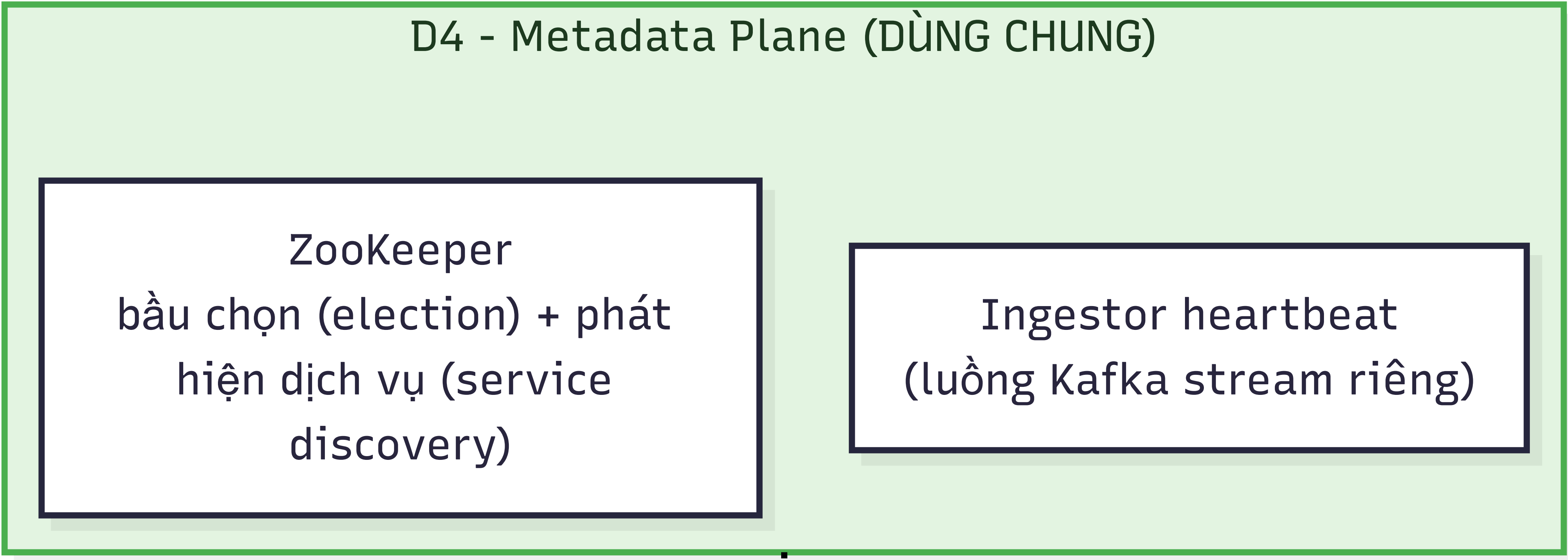
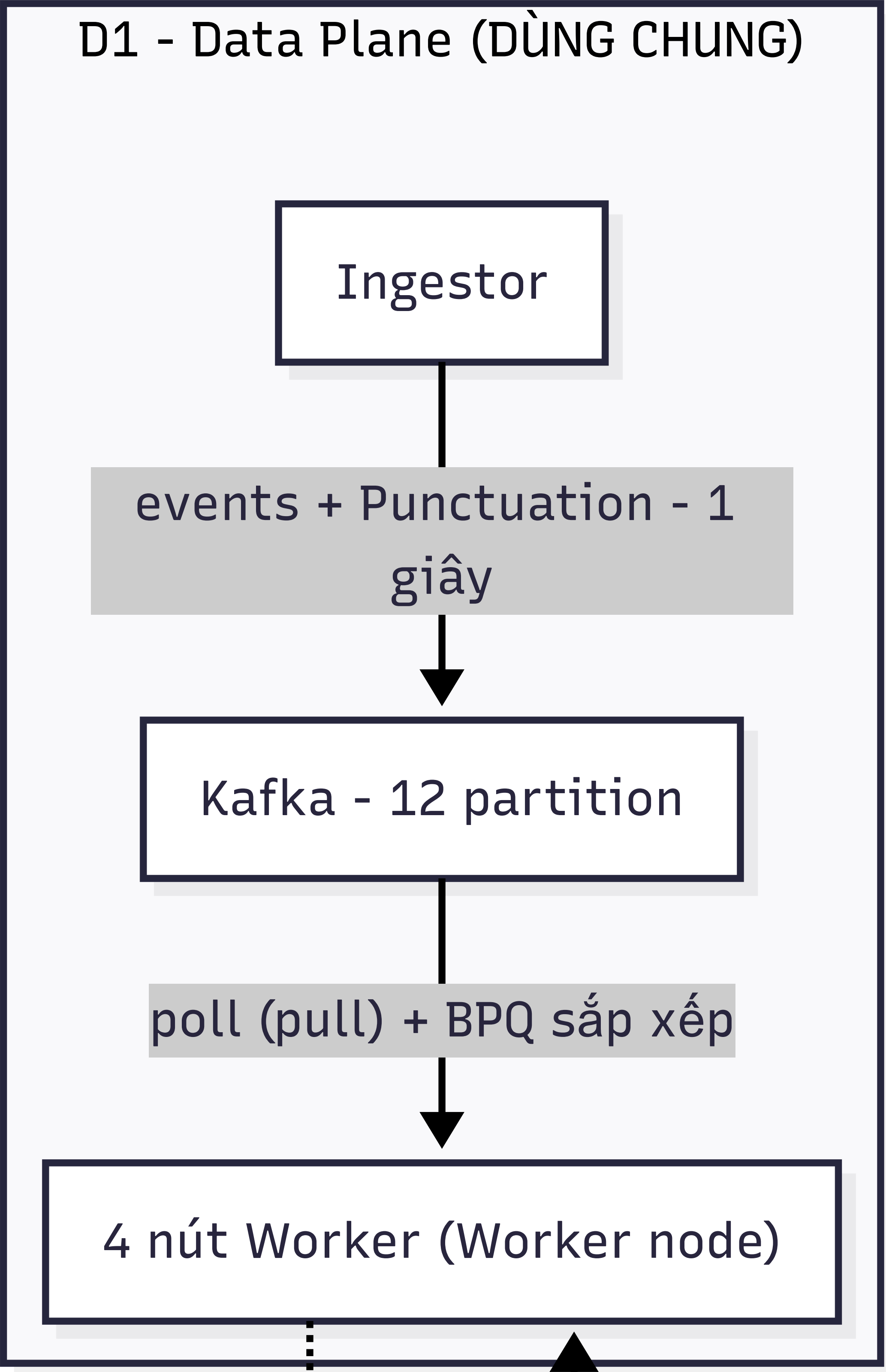




D1 - Data Plane (DÙNG CHUNG)

Ingestor

events + Punctuation - 1
giây

Kafka - 12 partition

poll (pull) + BPQ sắp xếp

4 nút Worker (Worker node)

ghi trạng thái (state)

D5 - State Plane (DÙNG CHUNG)

Tier1 - RocksDB (local)

checkpoint 10 giây

Tier2 - Phân vùng chia sẻ
(Shared Volume)

flush + DR 5 phút

Tier3 - MinIO

D4 - Metadata Plane (DÙNG CHUNG)

ZooKeeper
bầu chọn (election) + phát
hiện dịch vụ (service
discovery)

Ingestor heartbeat
(luồng Kafka stream riêng)

heartbeat / watermark

phát hiện (discovery) +
kiểm tra sức khỏe

Control Plane (HOÁN ĐỔI THEO CHẾ ĐỘ - MODE)

D2 - Strict Coordinator
(Raft 3 node)
 $W_{global} = \min(LW_i)$

D3 - Heuristic Aggregator
(Active-Standby)
 $W_{global_h} = \min(W_h) + DLQ$